

Lab: Planning Your Inventive Project

Objective

Develop a creative project plan for your invention that manages the exploration-exploitation trade-off, accounts for multiple types of uncertainty, defines stage-gate decision criteria, and matches planning depth to prediction reliability. By the end of this lab, you will have a structured but adaptive plan for advancing your invention from its current stage to its next major milestone.

Materials / Prerequisites

- Your own invention idea or project
- Paper and pen
- A timer
- Completed reading of Module 08: Planning the Creative Process

Part 1: Assess Your Current Position (10 minutes)

Step 1 (5 min): Where is your invention project right now? Describe its current state:

Current stage (circle one): Discovery / Scoping / Development / Validation / Launch

Write your response here

What do you know with confidence (low uncertainty)?

Write your response here

What do you know only roughly (moderate uncertainty)?

Write your response here

What do you not know at all (high uncertainty)?

Write your response here

Step 2 (5 min): Classify your uncertainties:

Uncertainty	Type (Parametric = values unknown / Structural = model unknown)	Impact if Wrong (High/Med/Low)	Can Be Tested? (Yes/No/How?)

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Part 2: Exploration-Exploitation Assessment (10 minutes)

The exploration-exploitation trade-off is the most important strategic decision in creative work. Explore too long and you never finish anything — you become a perpetual ideator with a drawer full of concepts. Converge too early and you miss better solutions — you become an efficient developer of mediocre ideas. The Active Inference framework provides guidance:

explore when the expected information gain from generating more alternatives exceeds the expected value of developing the current best option. Converge when further exploration is unlikely to discover a concept significantly better than the best one you already have. This assessment helps you determine which phase your project should currently be in.

Step 1 (5 min): Evaluate whether you should currently be exploring (generating alternatives) or exploiting (developing your current approach):

How many significantly different solution approaches have you considered?

Write your response here

Are your recent ideation sessions producing genuinely novel concepts, or variants of the same approach?

Write your response here

Do you have a concept that satisfies all critical constraints?

Write your response here

What is the cost of exploring for one more week versus committing now?

Write your response here

Step 2 (5 min): Based on your answers, should you be: - [] Exploring more (generating and evaluating alternative approaches) - [] Beginning to converge (narrowing from multiple approaches to one) - [] Fully committed (developing a single approach in detail)

Justify your assessment:

Write your response here

If you should be exploring more, what specific alternatives should you investigate?

Write your response here

If you should be converging, what criteria will you use to select the best approach?

Write your response here

Part 3: Design Your Stage-Gate Plan (15 minutes)

A stage-gate plan divides your inventive project into discrete stages separated by decision points (gates). Each stage involves creative work — building, testing, learning. Each gate

involves evaluation — do we have enough evidence to proceed, or should we pivot or stop? The power of stage-gate planning is that it makes implicit decisions explicit. Without gates, projects drift — continuing out of momentum rather than evidence, or pivoting out of frustration rather than analysis. With gates, every continuation is a conscious choice supported by evidence against defined criteria.

Step 1 (10 min): Define 3-4 stages for your invention project from its current state to a meaningful milestone (first functional prototype, first user test, first paying customer, etc.). For each stage, define the gate criteria:

Stage	Description	Duration Estimate	Key Activities	Gate Criteria (What must be true to proceed?)
Stage 1				

Write your response here

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Write your response here

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Write your response here

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Write your response here

|| Gate 1 decision | Continue / Pivot / Stop based on:

Write your response here

|| Stage 2 |

Write your response here

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|| Gate 2 decision | Continue / Pivot / Stop based on:

Write your response here

|| Stage 3 |

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|| Gate 3 decision | Continue / Pivot / Stop based on:

Write your response here

|| Stage 4 |

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Step 2 (5 min): For each gate, define what a "pivot" would look like. What would you do differently if the evidence at the gate does not support continuing the current approach?

Gate 1 pivot option:

Write your response here

Gate 2 pivot option:

Write your response here

Gate 3 pivot option:

Write your response here

Part 4: Information-Maximizing Milestones (10 minutes)

Traditional project planning orders activities by logical dependency — you cannot test before you build, you cannot build before you design. Information-maximizing planning adds a second ordering criterion: test the most dangerous assumptions first. A "dangerous" assumption is one that is both high-impact (if wrong, the project fails) and testable (you can design an experiment to check it). By testing these assumptions early, you either gain confidence that the project is viable or discover deal-breakers before investing significant resources. This is the "fail fast" principle applied at the planning level — structuring the project so that the most likely failure modes are encountered as early as possible.

Step 1 (5 min): Order your uncertainties from Part 1 by the product of (impact if wrong) x (testability). The highest-scoring uncertainties should be tested first:

Rank	Uncertainty	Impact x Testability Score	Test Designed to Resolve It
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Step 2 (5 min): Restructure your plan so that the highest-ranked uncertainty is tested in Stage 1, the second-highest in Stage 2, etc. Does this change the order of your original stage-gate plan?

Original order of activities:

Write your response here

Restructured (information-maximizing) order:

Write your response here

What changed and why:

Write your response here

Part 5: Multi-Horizon Planning (10 minutes)

Most planning fails not because the plans are wrong but because they apply the wrong level of detail to the wrong time horizon. A common mistake is to create a detailed 12-month plan — specifying exact activities, deliverables, and dates for every week of the next year. This plan will be wrong by month three because the structural uncertainty in creative work makes long-range prediction unreliable. An equally common mistake is to have no plan at all for the near term — living in vague aspirations without concrete next actions. Multi-horizon planning solves both problems by matching the level of detail to the reliability of prediction: precise for next week, directional for next quarter, visionary for next year.

Step 1 (5 min): Create plans at three different temporal horizons, with detail matched to prediction reliability:

Short horizon (next 2 weeks) — Specific actions:

Day/Week	Specific Action	Expected Output	What I Will Learn

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Medium horizon (next 2-3 months) — Objectives and milestones:

Write your response here

Long horizon (next 6-12 months) — Direction and vision:

Write your response here

Step 2 (5 min): When will you revise this plan? Set specific review dates:

Short-horizon review (plan specific actions for next 2 weeks):

Write your response here

Medium-horizon review (revise objectives based on accumulated evidence):

Write your response here

Long-horizon review (reassess overall direction):

Write your response here

Part 6: Risk and Adaptation (5 minutes)

Identify the three most likely ways your plan could fail, and describe your response to each:

Failure Mode	Probability (High/Med/Low)	Early Warning Signal	My Response
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Discussion / Debrief Questions

1. Was it difficult to classify your uncertainties as parametric versus structural? What does this difficulty tell you about the maturity of your generative model?
2. The exploration-exploitation assessment may have revealed that you are exploring too much or too little. Which tendency do you naturally exhibit, and how does it relate to your inventor archetype from Module 02?
3. Stage-gate processes force explicit go/no-go decisions. Is there a decision you have been avoiding making about your invention project? What evidence would help you make it?

4. Information-maximizing milestone ordering often produces a plan that is very different from the intuitive linear sequence. How did your restructured plan differ from your original, and which order makes more sense?
5. The multi-horizon planning approach asks you to be very detailed about the near future and very vague about the far future. Does this feel uncomfortable? How does matching planning depth to prediction reliability differ from how you normally plan?